

$\text{KNO}_3 \text{ and } \text{NaNO}_3 > \text{KNO}_2 > \text{HNO}_3$

FINAL TERM

Mod S

```
graph TD
    S1[S] --- L1[L]
    S1 --- S2[S]
    S2 --- L2[L]
    S2 --- S3[S]
    S3 --- L3[L]
```

<slnt>

18/12/24

Q (if mein else, apne if mein match karna chahiye)

$\langle \text{if} \rangle \rightarrow$ if exp then stnt |
if exp then stnt else stnt | other

if E_1 then if E_2 then S_1 else S_2
 $\hookrightarrow$ est $\hookrightarrow$

 $\langle I \rangle$

(lexeme mein tor kr
1. rna ha)

```

graph TD
    A["if exp then stat"] --> B["if"]
    A --> C["exp"]
    A --> D["then"]
    A --> E["stat"]
    E --> F["if"]
    E --> G["exp"]
    E --> H["then"]
    E --> I["else"]

```

Dangling it

X — X — X — X — X — X — X — X

<stdint>

5

exp

then

stnt

else

sent

7

1. ~~4~~

$$e_{xp}$$

then

stent

52

E 2

51

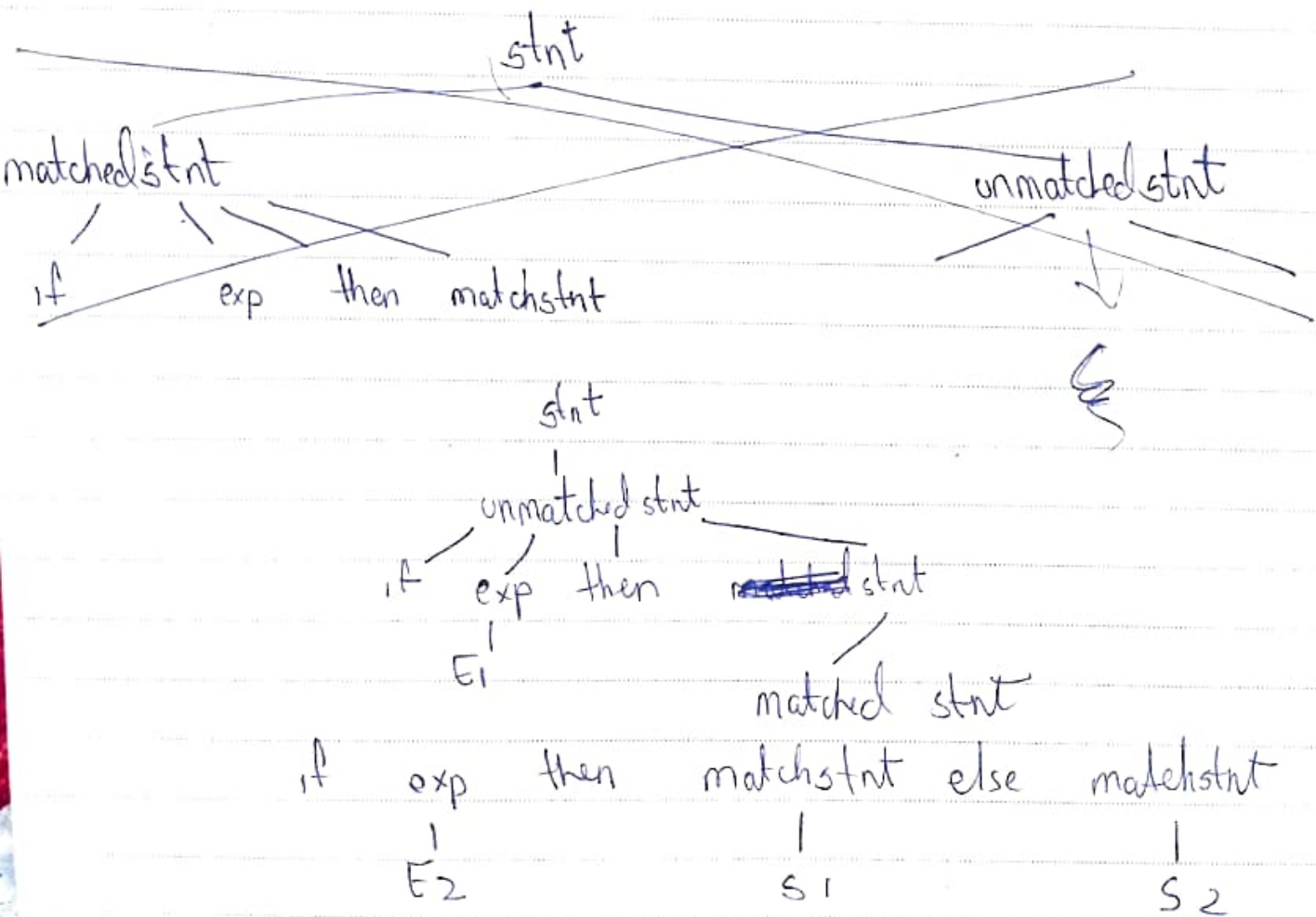
else should match
closest if ∞

stmt $\rightarrow$ matched stmt |
unmatched stmt

match stmt $\rightarrow$ if exp then match stmt else matched stmt | other

unmatched stmt $\rightarrow$ if exp then stmt |
if exp then matched stmt else unmatched stmt

if E_1 then E_2 then S_1 ~~the~~ else S_2



Date 3/1/25

Top down, Bottom Up

S

↓

String

↓

Starting Non Terminal

Call

$A \rightarrow AabAab$

↳ Immediate Left Recursion (No ϵ Terminal immediately calling itself)

To remove left recursion

$A \rightarrow A\alpha \mid B$

$A \rightarrow BA'$

$A' \rightarrow \alpha A' \mid \epsilon$

$\text{stmt} \rightarrow \langle \text{pr} \rangle \langle \text{stmt} \rangle \langle \text{else} \rangle \langle \text{stmt} \rangle$
Variability of Length $\rightarrow$ Recursive

$\beta \rightarrow$ Non Recursive Rule
 $\alpha \rightarrow$ Non Recursive part of Recursive Rule

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid id$

$E \rightarrow \underbrace{E}_A + \underbrace{T}_\alpha \mid \underbrace{T}_B$

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \epsilon$

$T \rightarrow FT'$

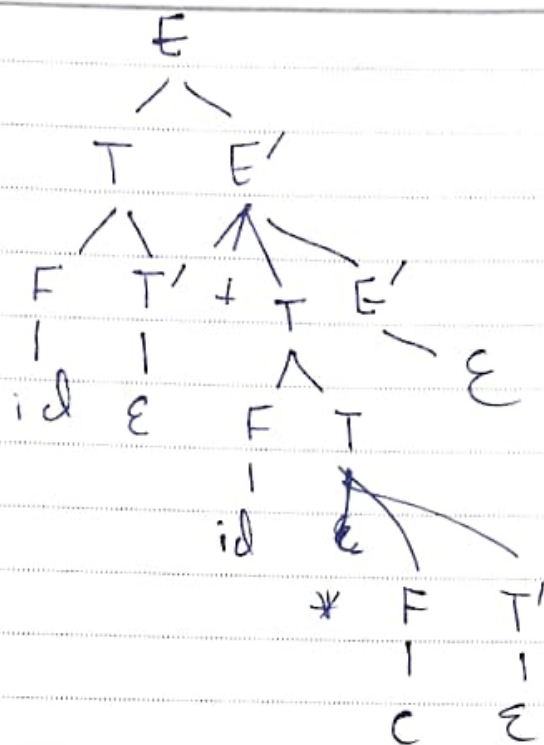
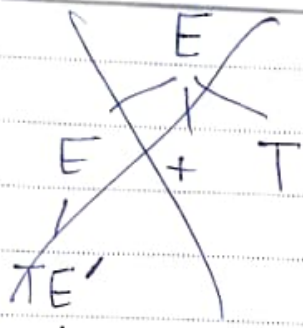
$T' \rightarrow *FT' \mid \epsilon$

~~$F \rightarrow *(E) \mid id$~~

$F \rightarrow (E) \mid id$

id+id*id

Date: _____



$A \rightarrow A, abcde, \alpha$

Scenario 1

$A \rightarrow A\alpha_1 B \mid A\alpha_2 B \mid A\alpha_3 B$ (Scenario 3)

$A \rightarrow BA'$

$A' \rightarrow \alpha_1 A' \mid \alpha_2 A' \mid \alpha_3 A' \mid \epsilon$

Scenario 2

$A \rightarrow A\alpha \mid B_1 \mid B_2 \mid B_3$

$A \rightarrow B_1 A' \mid B_2 A' \mid B_3 A'$

$A' \rightarrow \alpha A' \mid \epsilon$

$B \rightarrow$ Non Recursive

$B A'$ k sath

Remove Multiple Alpha's & Multiple Beta's

$S \rightarrow (L) \mid a$

$L \rightarrow L, S \mid S$

$L \rightarrow SL'$

$L \rightarrow , SL' \mid \epsilon$

Left Factoring

stmt $\rightarrow$ if expr then stmt, | if expr then stmt, else stmt,

stmt $\rightarrow$ E t S | E t S e S | a

stmt $\rightarrow$ E t S S'

S' $\rightarrow$ e S | e

Common Part remain same

uncommon derive from Non terminal

dec, print while do, switch
Loop, con, design VSL, grammar ambiguous

stmt $\rightarrow$ i E t S | i E t S e S | a

stmt $\rightarrow$ i E t S S'

S' $\rightarrow$ e I S e t S

(else if)

Complete all Construct

<start> $\rightarrow$ { <code> }
begin <stmt> end

<stmt> <de> <if else> & <loop>

<if else>

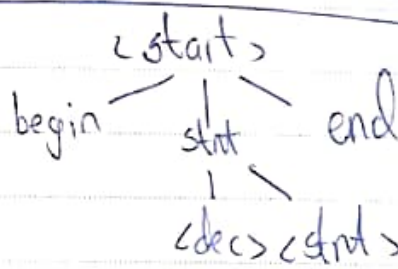
<dec> $\rightarrow$ <stmt>

<cond>

<print> $\rightarrow$ K

Connectivity Imp.

left rec $\rightarrow$ left fact

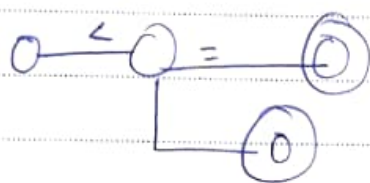


Date: 8/1/24

Recursive Descent Parser

$S \rightarrow cAd$
 $A \rightarrow ab|a$

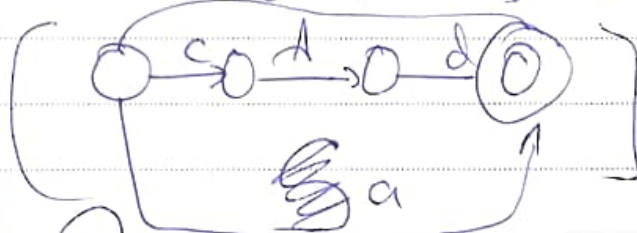
- Lexical Transition Diagram
 $\hookrightarrow$ To assign tokens



$\leq \rightarrow$ 1 token
 $\hookrightarrow$ 1 lexeme

$\hookrightarrow$ Designing them for Grammar

$S \rightarrow cAd|a|\epsilon|$
 ϵ 1 2 3 4



Representing S

Parsing $\rightarrow$ ① Top Down
 ② Bottom Up

Top Down

S

$\downarrow$

a

Bottom Up

abcd

$\downarrow$

S

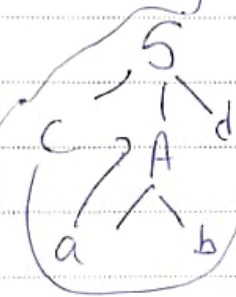
Back Tracking Structure

Start $\rightarrow$ ϵ $\rightarrow$ E $\rightarrow$ S
 $\rightarrow$ ϵ $\rightarrow$ E $\rightarrow$ ϵ $\rightarrow$ S

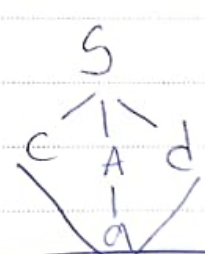
$S \rightarrow cAd$

$A \rightarrow ab|a$

cad



cabd
 we need
 cad



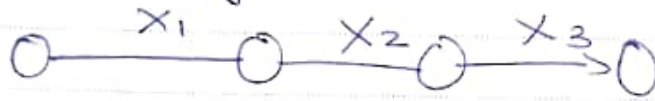
Code $\rightarrow$ Lexeme $\rightarrow$ Terminal

How to make Transition Diagram?

Eliminate Left Recursive + Left factor + Initial State + Final States $\rightarrow$ Connect Path from initial to final.
(Book Notation)

$A \rightarrow X_1 X_2 X_n$

Non Terminal k lie Initial & Final State



Transition $\rightarrow$ Some Non Terminal k lie Transition Diagram
Separate Diagram

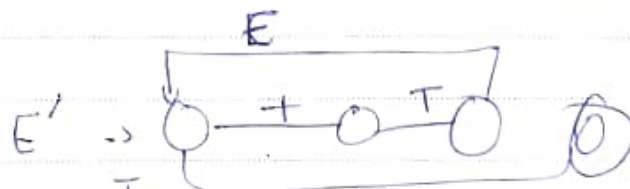
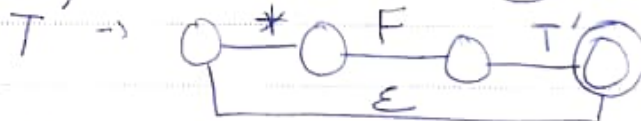
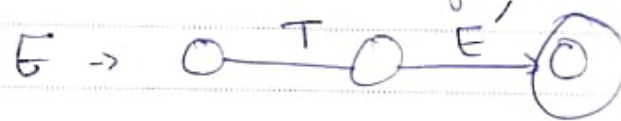
$E \rightarrow TE'$

$E' \rightarrow +TE' / \epsilon$

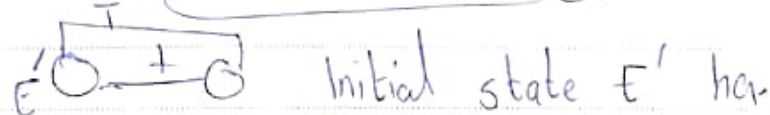
$T \rightarrow FT'$

$T' \rightarrow *FT' / \epsilon$

$F' \rightarrow (E) \text{ id}$



$L \rightarrow$



Date: _____

Non-Terminal	id	+ .	*	(	)	\$
E	$E \rightarrow TE'$			$E \rightarrow TE'$		
E'		$E' \rightarrow TE'$			$E' \rightarrow E$	$E' \rightarrow E$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow E$	$T' \rightarrow *FT'$			
F	$F \rightarrow id$			$F \rightarrow (E)$		

Stack

Input

Output

\$E	id + id * id \$	
\$E'T	id + id * id \$	$E \rightarrow TE'$
\$E'T'F	id + id * id \$	$T \rightarrow FT'$
\$E'T'id	id + id * id \$	$F \rightarrow id$
\$E'T'	+ id * id \$	$T \rightarrow E$
\$E'	+ id * id \$	
\$E'T	id * id \$	$E' \rightarrow TE'$
\$E'T'E	id * id \$	$T \rightarrow FT'$
\$E'T'id	id * id \$	$F \rightarrow id$
\$E'T'	* id \$	
\$E'T'F*	* id \$	$T' \rightarrow *FT$
\$E'T'F	id \$	
\$E'T'id	id \$	$F \rightarrow id$
\$E'T	\$	
\$E'	\$	$T' \rightarrow E$
\$	\$	$E' \rightarrow E$

$$\begin{aligned}
 E &\rightarrow TE' \\
 E' &\rightarrow +TE' \mid \epsilon \\
 T &\rightarrow FT' \\
 T' &\rightarrow *FT' \mid \epsilon \\
 F &\rightarrow (E)id
 \end{aligned}$$

funct for Table

First (A)
Follow (A)

$$\begin{aligned}
 \text{First}(F) &\rightarrow \{ (, id \} \\
 \text{First}(E) &\rightarrow \{ (, id \} \\
 \text{First}(T) &\rightarrow \{
 \end{aligned}$$

$$\begin{aligned}
 \text{First}(E) &= \text{First}(T) = \text{First}(EF) = \{ (, id \} \\
 \text{First}(E') &= \{ +, \epsilon \} \\
 \text{First}(T') &\rightarrow \{ *, \epsilon \}
 \end{aligned}$$

Follow

$$\begin{aligned}
 &\text{Follow}(E) \\
 \{ \$, > \} &\Rightarrow \cancel{\{ \epsilon, >, \$ \}}
 \end{aligned}$$

Non T ko Non T follow

Follow (T)

$$\{ +, >, \$ \}$$

\$ ko add

$$\begin{aligned}
 \text{Follow}(T) &= \text{first}(E') + \text{Follow}(E) \\
 &\{ +, >, \$ \}
 \end{aligned}$$

Date: _____

→ First of any Non Terminal is the first Terminal that is ~~not terminal~~ read from the Non Terminal.

→ First of any Non Ter → is Non Term → first NT ka first

15/1/25

$$\text{First}(E') = \{d, +, \epsilon\}$$

$$\text{First}(T') = \{d, *, \epsilon\}$$

$$\text{Follow}(T') = \text{Follow}(T) \text{ coz nothing is following } T'$$

$$\text{Follow}(F) = \{*, +, \epsilon, \$\}$$

$$\text{Follow}(E') = \{\$, \epsilon\}$$

LL(1)

	id	+	*	(	)	\$
E	$E \rightarrow TE'$			$E \rightarrow TE'$		
E'		$E' \rightarrow TE'$			$E \rightarrow \epsilon$	$E' \rightarrow \epsilon$
T	$T \rightarrow FT'$			$T \rightarrow FT'$		
T'		$T' \rightarrow \epsilon$	$T' \rightarrow *FT'$		$T' \rightarrow \epsilon$	$T' \rightarrow \epsilon$
F	$F \rightarrow id$					

Predictive Parsing Table

$\$ \rightarrow$ Starting N.T

Date: _____

First $\rightarrow$

- Follow $\rightarrow$
- ① Add $\$$ as starting Non Ter
 - ② If N.T(B) is following N.T.A, so the follow of A that Ter, the first of the following N.

~~A~~ N.T Follow(A) = First of (Following NT)

Follow(A) First of B

- ③ If first(B) includes NULL $\in$, then follow of its originator is includes too.
- $\rightarrow$ Nothing is following N.T $\rightarrow$ Originator

Q2)

$S \rightarrow iEtSS' \mid a$

$S' \rightarrow eS \mid \epsilon$

$E \rightarrow b$

First(S) $\rightarrow \{i, a\}$

First(S') $\rightarrow \{e, \epsilon\}$

First(E) $\rightarrow \{b\}$

Follow(S) $\rightarrow \{\$, e\}$

Follow(S') $\rightarrow \{e, \$\}$

Follow(E) $\rightarrow \{t\}$

Design P. Parsing Table

$XL L(1)X \downarrow$

Date: _____

	i	t	t	a	be	b b	\$
S	$S \rightarrow iEtSS'$			$S \rightarrow a$	$S \rightarrow e$		$S' \rightarrow \$$
S'					$S' \rightarrow eSt$		
E						$E \rightarrow b$	

→ 1 cell mein

1 NT ko doosra NT follow kr rha ha

2 → Ambiguous

$E \rightarrow$ follow of Originator

Row → Non Terminal

Column →

LL(1) Grammar

↳ left 2 right scanning
↳ left most derivation

(LL(K)) → Look ahead

LL(1) → Any Multi Define

BOTTOM UP PARSING

$S \rightarrow aABe$
 $A \rightarrow Abc \mid b$
 $B \rightarrow d$

Handle

$A \rightarrow b$
 $B \rightarrow d$

abbcde
aAbcde

aAde
aABe
 S

String

Starting Non Terminal

Called Bottom UP Approach

Right Most

$S \rightarrow$
 aABe
 aAde
 aAbcde
 abbcdde

reduce

Handles $\rightarrow$ Substring Matching $\rightarrow$ Right side of Rule

Ambiguous Nhi honi Chahiye Top down / Bottoms Up

Reduce
Shifting

Shift Reduce
Parsing.

Date: 17/1/25

Bottom $\rightarrow$ Shift Reduce Parse
 $\hookrightarrow$ LR Parser
 $\hookrightarrow$ LR(K) $\rightarrow$ Lookup
 SLR Parser

• k bad
 NT $\rightarrow$ uske rule
 same from I_0

$E' \rightarrow \cdot E$
 $E \rightarrow \cdot E + T$
 $E \rightarrow \cdot T$
 $T \rightarrow \cdot T * F$
 $T \rightarrow \cdot F$
 $F \rightarrow \cdot (E)$
 $F \rightarrow \cdot id$

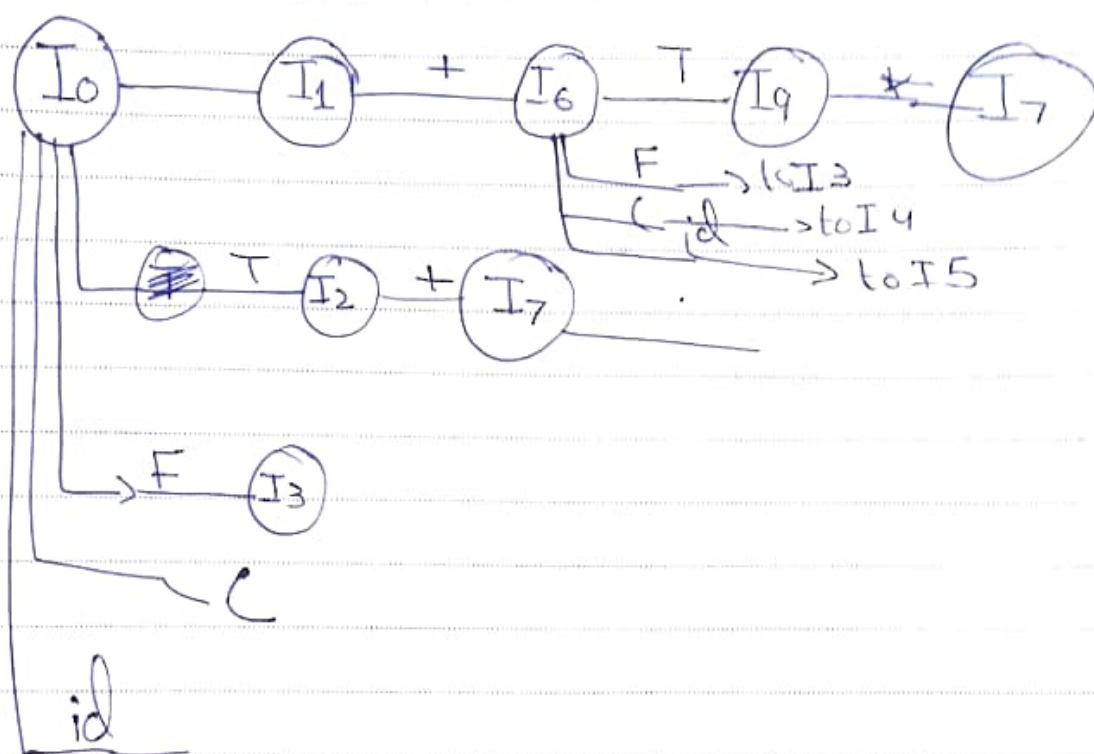
Goto (I_0, E)

$\left[\begin{array}{l} E' \rightarrow E. \\ E \rightarrow E. + T \end{array} \right] I_1$

goto (I_0, T)
 $\left[\begin{array}{l} E \rightarrow T. \\ T \rightarrow T. * F \end{array} \right] I_2$

goto (I_0, F)
 $T \rightarrow F.] I_3$

Go to ($I_0, ($)
 $F \rightarrow (\cdot E)$
 $E \rightarrow \cdot E + T$
 $E \rightarrow \cdot T$
 $T \rightarrow \cdot T * F$
 $T \rightarrow \cdot F$
 $F \rightarrow \cdot (E)$
 $F \rightarrow \cdot id$ } I_4


 $E \rightarrow E + T$
 $E \rightarrow T$
 $T \rightarrow T * F$
 $T \rightarrow F$
 $F \rightarrow (E)$
 $F \rightarrow id$

 First Follow (E) $\rightarrow$ (, id)

 First (T) $\rightarrow$ (, id)

 First (F) $\rightarrow$ (, id)

 Follow (E) $\rightarrow$ (, +, \$)

 Follow (T) $\rightarrow$ (*, +, >, \$)

 Follow (F) $\rightarrow$ (+, >, \$)

F ko koi follow nhi kr rahi, Originator or kr follow lkh deien

Go To Action Table

• baad mein aye Production Rule complete.

R4 → Reduce to Rule 4

F → id id ki jagah F

Simple LR Table

Shift Reduce

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Semantic Analysis

Syntax Directed
Translation

Syntax Directed
Definition

Translation
Scheme

Input → Parser → SDT → Evaluation Order

S. Directed

Date: _____

Production

$L \rightarrow E_n$

$E \rightarrow E_1 + T$

$E \rightarrow T$

$T \rightarrow T * F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow \text{digit}$

Semantic Rule

Print (E.val)

$E.val = E_1.val + T.val$

$E.val = T.val$

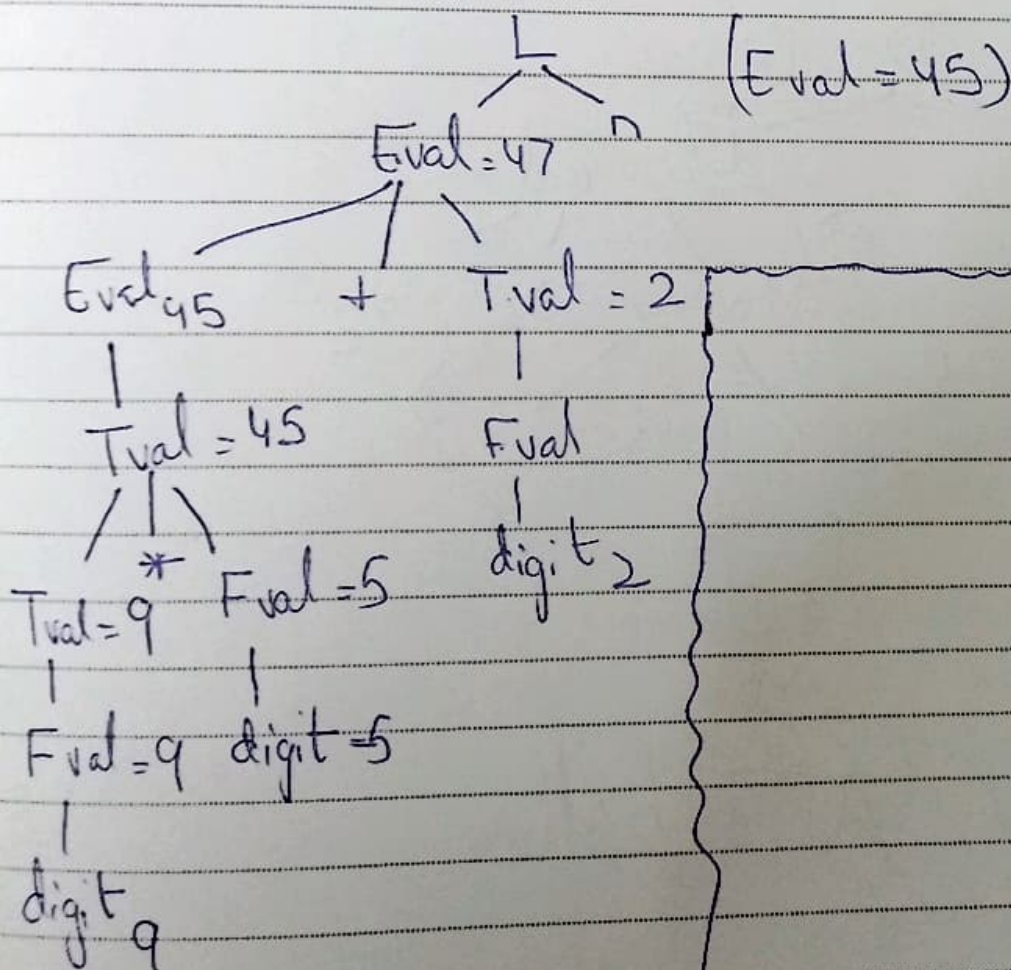
$T.val = T_1.val * F.val$

$T.val = F.val$

$F.val = E.val$

$F.val = \text{digit} / \text{ex.val}$

$9 \times 5 + 2$



Dependency Graph, evaluation Order

Inherited attribute

Date: _____

Product

$D \rightarrow TL$

$T \rightarrow \text{int}$

$T \rightarrow \text{real}$

$L \rightarrow L_1 \text{ in } id$

~~$L \rightarrow id$~~

$L \rightarrow id$

Semantic Rule

$L.in = T.type$

$T.type = \text{integer}$

$T.type = \text{real}$

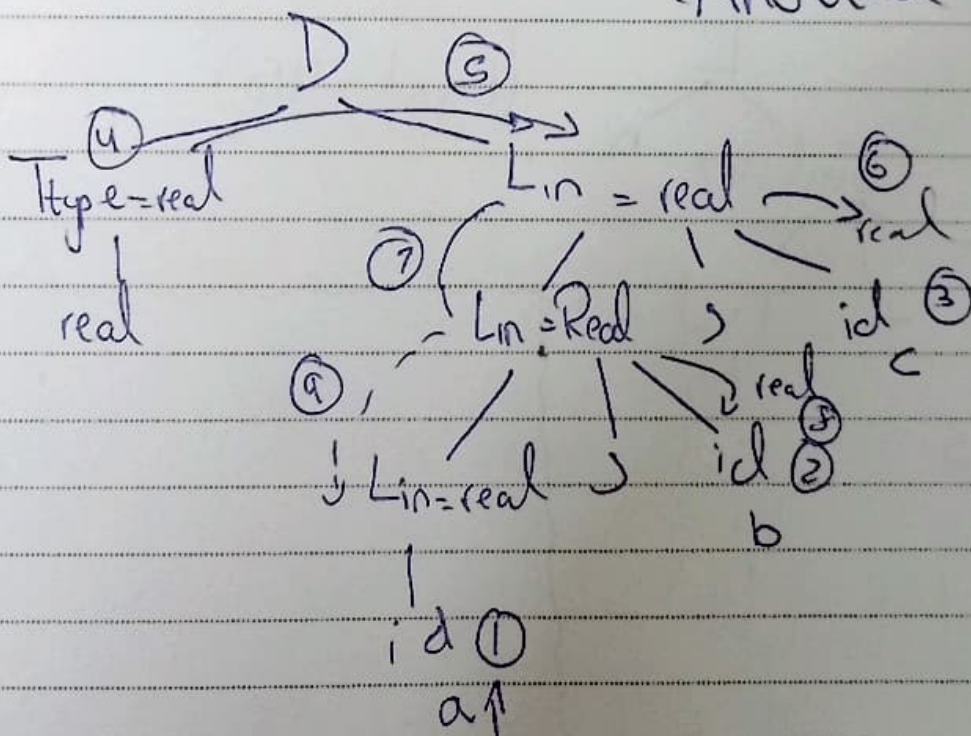
$L, L.in$

'add type (ident)'
 $L.in$

add type (ident)
 $L.in$

real a, b, c;

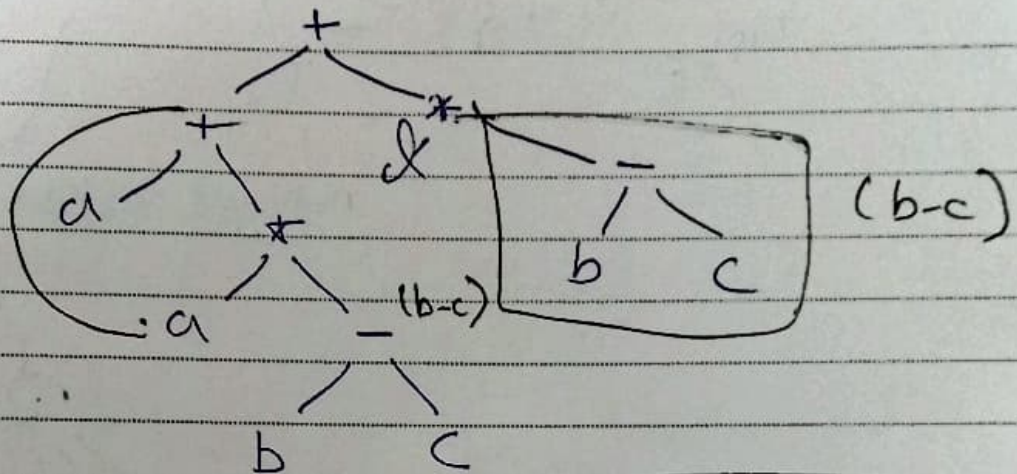
Annotated Parse



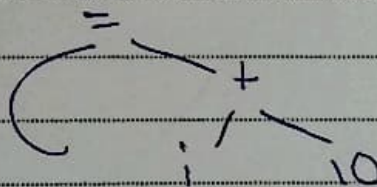
Inherited Attribute

Acyclic Graph

$$a + a * (b - c) + d * (b - c)$$



$$i = i + 10$$



Common 10
Reuse

Translation Scheme

$$E \rightarrow TR$$

$$R \rightarrow \text{addop}$$

$$T \rightarrow \text{num}$$

$$T \rightarrow \text{Print (addop)} \quad \text{or} \quad \text{Print (num.val)}$$

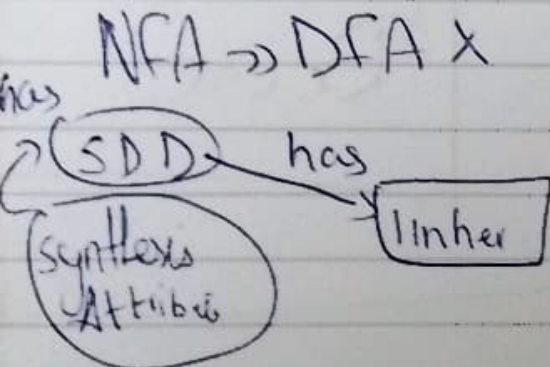
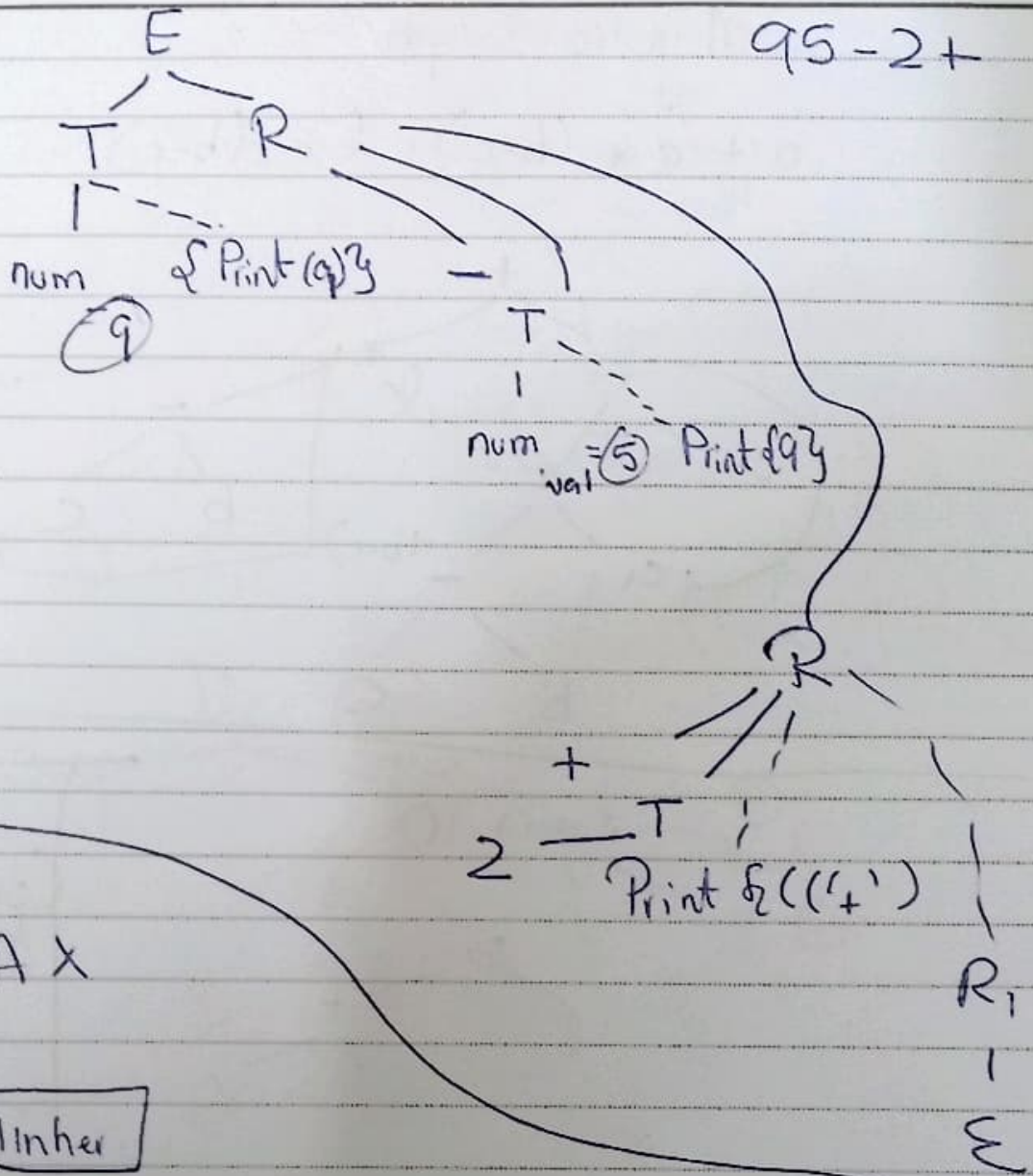
$$R, E$$

$$a - 5 + 2$$

Postfix Exp

Date: _____

95-2+



Phases, Syntax Top, Bottom SLR is bottom up
↳ LR

Theory Chp1

Chp1, lexical, Syntax
Left fact, Left Ref